

Ryan Anselm

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Education	The University of Texas at Austin Ph.D. in Computer Science Advised by Scott Aaronson and Nick Hunter-Jones NSF Graduate Research Fellow Columbia University B.S. in Computer Science, <i>Magna Cum Laude</i> Minors in Applied Math & Applied Physics GPA: 4.05/4.00	Sept. 2025 - Present Sept. 2021 - May 2024
Papers	Ryan Anselm, Michelle Ding, Dar Gilboa, and Sabee Grewal. “Optimal and Verifiable Quantum Advantages in Communication Complexity” . <i>Preprint</i> . Ryan Anselm, Srijita Kundu, Olivier Lalonde, and Ashwin Nayak. “An exponential separation between entanglement-assisted and unassisted one-way quantum communication” . <i>Preprint</i> .	
Presentations	<i>A Near-Maximal, Verifiable Quantum Advantage in One-Way Communication</i> . Invited Talk. Visa Research, August 2026. <i>Quantum-inspired Tree Tensor Network Methods for Compression and Transformation of Multivariate Functions</i> . R. Anselm, R. Levy, J. Tindall. Tensor Networks and Classical Optimization Session, American Physical Society Global Physics Summit, March 2025. <i>Machine learning optimal prognostic moments for single-category cloud microphysics parameterizations</i> . R. Anselm, M. van Lier-Walqui, K. Lamb. Processes of (Sub) Cloud Scales: Modeling, Observations, and Parameterizations Session, American Geophysical Union Fall Meeting, December 2022.	
Research Experience	Flatiron Institute , <i>Summer Research Assistant</i> Hosted by the Center for Computational Quantum Physics Supervised by Ryan Levy & Joseph Tindall - Contributed to a Julia library generalizing the Quantized Tensor Train (QTT) format to arbitrary Tree Tensor Networks (TTNs), a framework for compressed, low-rank representations of multivariate functions. - Ran large-scale numerical experiments on compute clusters to study function structure vs. optimal tree layout, and implemented efficient TTN operators (e.g., Discrete Fourier Transform, Koopman) for solving differential equations. Columbia Quantum Computing Theory Group Advised by Henry Yuen. Learning the Earth with AI and Physics (LEAP) REU Supervised by Kara Lamb, Marcus van Lier-Walqui - Trained neural network models to emulate cloud microphysics, achieving comparable accuracy to traditional bulk/bin simulation methods. Worked with the ML Python ecosystem (Numpy, Pytorch, Scikit-learn, Scipy, Xarray, etc.) to implement and evaluate models.	May 2024 - Aug. 2024 Aug. 2023 - May 2024 Jun. 2022 - Aug. 2022
Awards	2025 NSF Graduate Research Fellowship 2024 MIT iQuHack Quantum Hackathon, 1st place in track 2022 Top 350 placement, William Lowell Putnam Mathematical Competition	
Work Experience	Salesforce Associate Member of Technical Staff Salesforce Software Engineering Intern	Sept. 2024 - Nov. 2025 May 2023 - Aug. 2023

Teaching Experience	UT Austin CS358H Intro to Quantum Information Science, <i>Teaching Assistant</i>	Fall 2026
	Columbia University COMS 4232 Advanced Algorithms, <i>Teaching Assistant</i>	Spring 2024
	Columbia University COMS 4236 Quantum Computing, <i>Teaching Assistant</i>	Fall 2023
	Columbia University COMS 3261 Computer Science Theory, <i>Teaching Assistant</i>	Spring 2023
	Columbia University CSOR 4231 Analysis of Algorithms I, <i>Teaching Assistant</i>	Fall 2022
Outreach & Service	Columbia University Science Olympiad, <i>Co-President & Co-Founder</i>	2021 - 2024
	- Co-founded an undergraduate student group and led a team of 12 undergraduates to organize an annual Science Olympiad invitational tournament hosted at Columbia University for 500+ high school students. - Authored 12+ Science Olympiad tests as an volunteer for several tournaments including the 2021 national tournament, Columbia, MIT, UT Austin, UCincinnati, and UPenn invitationals.	
	Columbia University Undergraduate Learning Seminar in Theoretical Computer Science	2022-2024
	Organized a reading group on quantum complexity theory and presented on foundational papers.	